LANDOS ATLAS by GIGI LABS

Cinematic Implementation Pack

Exact Route Map, GSAP Timeline Breakdown, R 3 F Object List, Asset Production Brief, Scene Logic, and Codex Execution Prompt

---

1 . Purpose

This pack continues the LandOS Atlas Cinematic Experience System and translates it into an implementation-ready build document for Codex, design agents, 3 D artists, and motion engineers.

It contains: - exact cinematic route strategy - exact scene route/page map - exact GSAP timeline by section - exact React Three Fiber object inventory - exact asset production brief - exact co hook map - exact motion state model - exact mobile fallback plan - exact implementation comm for Codex

This is the execution layer for the immersive storytelling surface of LandOS Atlas.

---

2 . Cinematic Product Surface Strategy

LandOS Atlas should have a dedicated cinematic layer separate from the core app workspace.

Recommended Route Families

Public Marketing / Storytelling Routes

- `/`
- `/experience`
- `/platform`
- `/for-developers`
- `/for-investors`
- `/for-governments`
- `/for-enterprises`

App Routes

- `/app`
- `/app/projects/:id`
- `/app/scenarios`

- `/app/feasibility`
- `/app/billing`

Rule

The deepest cinematic motion should live on: - `/` - `/experience` - `/platform`

The audience-specific pages should reuse lighter cinematic patterns.

The app should remain premium and elegant, but not fully scroll-cinematic.

---

3. Exact Cinematic Route Map

Route 1 — `/`

Purpose

High-conversion flagship homepage.

Experience Level

Moderate-to-high cinematic.

Sections

1. Hero planetary intro
2. Parcel discovery reveal
3. Scenario generation highlight
4. Feasibility highlight
5. Collaboration highlight
6. Platform ecosystem highlight
7. CTA

Route 2 — `/experience`

Purpose

Full immersive scroll masterpiece.

Experience Level

Maximum cinematic depth.

Sections

1. Enter the planetary OS
2. Zoom into land
3. Reveal parcel intelligence
4. Scenario deconstruction
5. Optimization engine sequence
6. Financial core reveal
7. Collaboration network sequence
8. Planetary closing sequence

This is the flagship narrative route.

Route 3 — `/platform`

Purpose

Structured premium overview with cinematic modules.

Experience Level

Moderate cinematic with clearer product explanation.

Sections

1. platform intro
2. modules grid
3. intelligence engine
4. finance engine
5. collaboration engine
6. enterprise and government surfaces
7. CTA

Route 4 — Audience Pages

Purpose

Industry-specific storytelling with lighter motion.

Pages

- `/for-developers`
- `/for-investors`
- `/for-governments`

- `/for-enterprises`

Experience Level

Light-to-moderate cinematic.

Use: - parallax hero - glass panels - subtle section reveals - smaller data-driven scene animations

---

4 . Exact `/experience` Page Structure

```
ExperiencePage
  CinematicShell
    SmoothScrollProvider
    SceneProgressProvider
    HeroPlanetarySection
    ParcelDiscoverySection
    IntelligenceRevealSection
    ScenarioDeconstructionSection
    OptimizationOrbitSection
    FinancialCoreSection
    CollaborationNetworkSection
    PlanetaryClosingSection
```

Each Scene Should Contain

- pinned story wrapper
- R 3 F or visual canvas region
- text overlay system
- metrics overlay system
- progress-bound animation timeline
- fallback states

---

5 . Exact Scene-by-Scene GSAP Timeline Breakdown

Scene 1 — HeroPlanetarySection

Goal

Introduce LandOS Atlas as world-scale infrastructure.

Visuals

- dark atmospheric background
- animated globe or world map surface
- faint moving particles
- title and subtitle
- CTA panel

Timeline Events

1. fade in ambient background
2. globe slowly rotates into view
3. heading rises and sharpens from blur
4. subtitle fades in
5. CTA panel slides upward with glow edge
6. background particles intensify slightly

Timeline Structure

```
0.00 atmosphere opacity 0 → 1
0.20 globe scale 0.92 → 1, rotation Y subtle
0.40 title y 40 → 0, opacity 0 → 1
0.55 subtitle opacity 0 → 1
0.70 CTA panel y 30 → 0, opacity 0 → 1
0.85 ambient glow intensity increases slightly
```

Scene 2 — ParcelDiscoverySection

Goal

Move from global scale to a specific parcel.

Visuals

- map zoom
- search marker or pulse
- parcel boundary trace
- floating land metrics

Timeline Events

1. camera pushes into map region
2. pulse marker locks target area
3. parcel boundary draws itself

4. area/perimeter cards slide in
5. region labels and intelligence indicators appear

Timeline Structure

```
0.00 camera z push-in
0.20 target pulse expands and contracts
0.35 parcel boundary line draws
0.55 metric cards left/right enter
0.75 parcel glow edge intensifies
0.90 map context dims around selected parcel
```

Scene 3 — IntelligenceRevealSection

Goal

Reveal what the platform understands about land.

Visuals

- parcel prism
- terrain mesh
- access lines
- context nodes
- glass intelligence cards

Timeline Events

1. parcel extrudes upward from map plane
2. terrain mesh rises subtly underneath
3. access vectors illuminate
4. context points orbit softly
5. cards animate into place with stagger

Timeline Structure

```
0.00 parcel extrusion height 0 → target
0.20 terrain mesh opacity 0 → 1
0.35 access line illumination sweeps in
0.55 context nodes scale/opacity in
0.70 glass cards stagger upward
0.90 recommendation pulse begins subtly
```

Scene 4 — ScenarioDeconstructionSection

Goal

Show parcel transforming into development scenarios.

Visuals

- parcel block
- roads layer
- plot cells
- reserve zones
- labels in space

Timeline Events

1. parcel tilts slightly toward camera
2. roads separate downward
3. plots split upward into cells
4. reserves rise as translucent colored volumes
5. labels and metrics lock to components
6. all layers hover in controlled alignment

Timeline Structure

```
0.00 parcel tilt + scale emphasis
0.20 roads y +80, opacity 0 → 1
0.40 plots y -60 with stagger, opacity 0 → 1
0.60 reserves y -120, opacity 0 → 1
0.75 labels fade and pin in
0.90 layer glows and subtle hover state begin
```

Reverse Scroll Requirement

All components must settle back into one integrated parcel stack smoothly.

Scene 5 — OptimizationOrbitSection

Goal

Show multiple scenario options and recommendation logic.

Visuals

- several scenario objects
- orbit or fan layout
- score bars
- recommendation halo
- comparison panel

Timeline Events

1. primary scenario shifts center
2. alternate scenarios fan outward or orbit
3. each scenario receives metric tags
4. recommendation halo locks to best option
5. score bars animate upward
6. comparison panel fades in

Timeline Structure

```
0.00 center object stabilizes
0.20 option A/B/C spread outward
0.45 metric labels appear per option
0.60 recommendation ring forms around selected option
0.75 score bars animate from 0 to final values
0.90 comparison panel fades up
```

Scene 6 — FinancialCoreSection

Goal

Reveal the economic intelligence inside the plan.

Visuals

- scenario shell
- peeling outer geometry
- revenue ring
- cost layer
- margin core
- animated stats and charts

Timeline Events

1. outer shell opens/peels away

2. revenue ring rotates into view
3. cost layer reveals beneath
4. margin core pulses
5. values count up
6. investor summary panel appears

Timeline Structure

```
0.00 outer shell split open
0.25 revenue ring rotate/scale in
0.45 cost layer rise/opacity in
0.65 margin core pulse + glow
0.78 numbers count up
0.88 investor panel fades in
```

Scene 7 — CollaborationNetworkSection

Goal

Show that LandOS Atlas coordinates the ecosystem.

Visuals

- floating network graph
- stakeholder nodes
- task/comment/approval cards
- directional links

Timeline Events

1. central project node appears
2. developer/investor/planner/gov nodes animate in
3. lines connect in sequence
4. cards for comment/task/approval appear
5. network stabilizes as operating system view

Timeline Structure

```
0.00 central node appears
0.20 outer nodes scale/opacity in
0.40 lines draw between nodes
0.60 task/comment/approval cards float in
```

```
0.80 network pulsing rhythm settles
0.92 section text locks in
```

Scene 8 — PlanetaryClosingSection

Goal

End with scale and inevitability.

Visuals

- zoom out from system to planetary grid
- multiple glowing parcels across world
- final headline
- CTA

Timeline Events

1. collaboration network recedes
2. camera lifts to regional then global view
3. glowing activity nodes appear across map
4. final title fades in
5. CTA locks center with premium pulse

Timeline Structure

```
0.00 network fades and compresses
0.20 camera zooms outward
0.45 world-level nodes illuminate
0.65 final title enters
0.82 CTA enters with subtle glow pulse
0.95 background settles into low-motion state
```

---

6. Exact Scene State Model

Each scene should have a normalized `progress` value from `0` to `1`.

Recommended State Interface

```
export type SceneProgress = {
  hero: number;
  parcelDiscovery: number;
  intelligenceReveal: number;
  scenarioDeconstruction: number;
  optimizationOrbit: number;
  financialCore: number;
  collaborationNetwork: number;
  planetaryClosing: number;
};
```

Usage

- R 3 F objects consume progress to adjust transforms/materials
- UI overlays consume progress to animate opacity, blur, y-offset
- color system consumes global story phase to shift gradients/glow

---

7 . Exact React Three Fiber Object List

Core Story Objects

7 . 1 GlobeObject

Purpose: establish planetary scale.

7 . 2 WorldMapSurface

Purpose: show global intelligence plane.

7 . 3 ParcelPrism

Purpose: central parcel hero object.

7 . 4 BoundaryLines

Purpose: traced parcel edges.

7 . 5 TerrainMesh

Purpose: terrain/context suggestion layer.

7.6 AccessVectors

Purpose: road and mobility intelligence.

7.7 PlotCells

Purpose: plot blocks that can separate and reassemble.

7.8 RoadLayer

Purpose: internal road network object.

7.9 ReserveVolumes

Purpose: translucent amenity/open-space components.

7.10 ScenarioOrbitGroup

Purpose: group of alternate scenario models.

7.11 RecommendationHalo

Purpose: selected scenario emphasis.

7.12 RevenueRing

Purpose: financial reveal object.

7.13 CostLayer

Purpose: cost representation object.

7.14 MarginCore

Purpose: profit/margin emphasis object.

7.15 CollaborationNodeGraph

Purpose: multi-stakeholder network visualization.

7.16 PlanetarySignalField

Purpose: ending scene world-scale intelligence nodes.

---

8 . Exact R 3 F Scene File Map

```
apps/web/src/components/r3f/
  AtlasCanvas.tsx
  AtlasExperienceScene.tsx
  AtlasCameraRig.tsx
  scene-objects/
    GlobeObject.tsx
    WorldMapSurface.tsx
    ParcelPrism.tsx
    BoundaryLines.tsx
    TerrainMesh.tsx
    AccessVectors.tsx
    PlotCells.tsx
    RoadLayer.tsx
    ReserveVolumes.tsx
    ScenarioOrbitGroup.tsx
    RecommendationHalo.tsx
    RevenueRing.tsx
    CostLayer.tsx
    MarginCore.tsx
    CollaborationNodeGraph.tsx
    PlanetarySignalField.tsx
  materials/
    GlowMaterial.tsx
    GlassMaterial.tsx
    WireframeMaterial.tsx
  hooks/
    useSceneProgress.ts
    useResponsiveSceneMode.ts
```

---

9 . Exact Cinematic Hook Map

```
apps/web/src/hooks/cinematic/
  useLenisScroll.ts
  useSceneProgressStore.ts
  usePinnedSceneTimeline.ts
  useColorShiftSystem.ts
  useReducedMotionMode.ts
  useDevicePerformanceTier.ts
```

Hook Responsibilities

`useLenisScroll`

Initialize and sync smooth scrolling.

`useSceneProgressStore`

Expose scene progress globally.

`usePinnedSceneTimeline`

Create modular GSAP timelines for sections.

`useColorShiftSystem`

Shift CSS variables and glow states by scroll phase.

`useReducedMotionMode`

Respect reduced-motion preferences.

`useDevicePerformanceTier`

Decide heavy/light scene behavior.

---

10. Exact Asset Production Brief

A design/3D team should prepare the following assets.

10.1 3D Models

Required

- `globe.glb`
- `parcel-prism.glb`
- `scenario-stack.glb`
- `revenue-ring.glb`
- `financial-core.glb`
- `node-network.glb`

Style Rules

- low-to-mid poly

- clean futuristic geometry
- no unnecessary material complexity
- suitable for glow/rim-light scenes
- designed to animate into parts if needed

10.2 Textures

Required

- atmospheric gradient texture
- topo grid texture
- noise texture
- subtle scanline texture
- halo glow texture
- world signal texture

10.3 Motion Graphics / Fallback Assets

Required

- mobile hero fallback video
- lightweight looping map ambient video
- poster images for lazy-loaded scenes

10.4 UI Visual Assets

Required

- glass panel background treatment
- icon system for intelligence, finance, collaboration, governance, infrastructure
- premium section dividers

---

11. Exact Asset Folder Structure

```
public/cinematic/
  models/
    globe.glb
    parcel-prism.glb
    scenario-stack.glb
    revenue-ring.glb
    financial-core.glb
    collaboration-network.glb
  textures/
```

```
    atmosphere-gradient.webp
    topo-grid.webp
    noise.png
    glow-ring.webp
    world-signals.webp
  videos/
    hero-mobile-fallback.mp4
    parcel-scene-mobile.mp4
  posters/
    hero-poster.webp
    parcel-poster.webp
    finance-poster.webp
```

---

12. Exact Component Architecture for the Cinematic Route

```
ExperiencePage
  CinematicPageShell
    SmoothScrollProvider
    ScrollGradientField
    AmbientParticles
    HeroPlanetarySection
      HeroTextOverlay
      HeroCTA
      HeroGlobeCanvas
    ParcelDiscoverySection
      DiscoveryTextOverlay
      DiscoveryMetricsOverlay
      ParcelCanvas
    IntelligenceRevealSection
      IntelligenceCardsOverlay
      TerrainCanvas
    ScenarioDeconstructionSection
      ScenarioTextOverlay
      ScenarioMetricsOverlay
      ScenarioCanvas
    OptimizationOrbitSection
      OptimizationTextOverlay
      ComparisonPanel
      OrbitCanvas
    FinancialCoreSection
      FinancialTextOverlay
      FinanceMetricsOverlay
```

```
      FinancialCanvas
    CollaborationNetworkSection
      CollaborationTextOverlay
      StakeholderCardsOverlay
      NetworkCanvas
    PlanetaryClosingSection
      ClosingTextOverlay
      FinalCTA
      ClosingCanvas
```

---

13. Exact Color and CSS Variable System

Use CSS variables driven by scene progress.

Suggested Variables

```
:root {
  --atlas-bg: #05070B;
  --atlas-panel: rgba(255,255,255,0.08);
  --atlas-blue: #1397FF;
  --atlas-cyan: #4FD1FF;
  --atlas-gold: #F4C542;
  --atlas-white: #EAF6FF;
  --atlas-glow-strength: 0.4;
  --atlas-scene-hue-shift: 0deg;
}
```

Scroll-Reactive Changes

- hero: blue/cyan emphasis
- optimization: cyan/gold recommendation emphasis
- finance: gold/green accents
- collaboration: blue/teal network emphasis
- closing: planetary blue with gold CTA focus

---

14. Exact Mobile Fallback Rules

If mobile or low-performance tier

Replace or simplify: - reduce live particle count - remove some bloom/post-processing - shorten sections - replace heavy 3D scenes with pre-rendered video or simplified meshes - limit multi-orbit scenes - keep the story but lighten the rendering

Mobile Scene Strategy

- Hero still interactive, but simpler
- Parcel scene may use video or 2.5D layered transform
- Optimization and finance scenes can blend live UI + lighter canvas

---

15. Exact Performance Tiers

Tier 1 — High-End Desktop

- full R3F scenes
- high particle count
- bloom allowed
- full pinned scroll sequences

Tier 2 — Mid Device

- reduced particles
- reduced postprocessing
- simplified materials
- shorter pinned durations

Tier 3 — Mobile / Low-End

- critical scenes only in full 3D
- video/canvas fallback for heavy sections
- reduced parallax depth complexity
- no unnecessary shader cost

---

16. Exact GSAP/ScrollTrigger Hook Pattern

```
'use client';

import { useLayoutEffect } from 'react';
import gsap from 'gsap';
import ScrollTrigger from 'gsap/ScrollTrigger';

gsap.registerPlugin(ScrollTrigger);

export function usePinnedSceneTimeline(
  sectionRef: React.RefObject<HTMLElement>,
  buildTimeline: (tl: gsap.core.Timeline) => void,
  end = '+=2000'
) {
  useLayoutEffect(() => {
    if (!sectionRef.current) return;

    const ctx = gsap.context(() => {
      const tl = gsap.timeline({
        scrollTrigger: {
          trigger: sectionRef.current,
          start: 'top top',
          end,
          scrub: true,
          pin: true,
          anticipatePin: 1,
        },
      });

      buildTimeline(tl);
    }, sectionRef);

    return () => ctx.revert();
  }, [sectionRef, buildTimeline, end]);
}
```

---

17. Exact Example R3F Progress Wiring

```
'use client';

import { useMemo } from 'react';
```

```
import { Canvas } from '@react-three/fiber';
import { ParcelPrism } from './scene-objects/ParcelPrism';
import { PlotCells } from './scene-objects/PlotCells';
import { RoadLayer } from './scene-objects/RoadLayer';
import { ReserveVolumes } from './scene-objects/ReserveVolumes';

export function ScenarioCanvas({ progress }: { progress: number }) {
  const plotSpread = useMemo(() => progress, [progress]);

  return (
    <Canvas camera={{ position: [0, 3, 8], fov: 36 }}>
      <ambientLight intensity={0.5} />
      <directionalLight intensity={1.8} position={[3, 5, 4]} />
      <ParcelPrism progress={progress} />
      <RoadLayer progress={progress} />
      <PlotCells progress={plotSpread} />
      <ReserveVolumes progress={progress} />
    </Canvas>
  );
}
```

---

18. Exact Implementation Order for the Cinematic Surface

Step 1

Create cinematic route shell and smooth scroll provider.

Step 2

Build HeroPlanetarySection.

Step 3

Build ParcelDiscoverySection.

Step 4

Build IntelligenceRevealSection.

Step 5

Build ScenarioDeconstructionSection.

Step 6

Build OptimizationOrbitSection.

Step 7

Build FinancialCoreSection.

Step 8

Build CollaborationNetworkSection.

Step 9

Build PlanetaryClosingSection.

Step 10

Tune performance tiers and mobile fallbacks.

---

19. Exact Codex Prompt for Cinematic Build

```
Build the LandOS Atlas cinematic storytelling surface.

Use Next.js / React with:
- GSAP + ScrollTrigger
- Lenis
- React Three Fiber / Three.js
- Framer Motion
- Tailwind CSS

Create the following routes and experiences:
- /
- /experience
- /platform
- /for-developers
- /for-investors
- /for-governments
```

```
- /for-enterprises

The flagship cinematic route is /experience.

Implement the following pinned scroll sections in order:
1. HeroPlanetarySection
2. ParcelDiscoverySection
3. IntelligenceRevealSection
4. ScenarioDeconstructionSection
5. OptimizationOrbitSection
6. FinancialCoreSection
7. CollaborationNetworkSection
8. PlanetaryClosingSection

Requirements:
- scroll-scrubbed GSAP timelines
- reverse-safe animation on upward scroll
- layered parallax depth
- premium glass UI overlays
- R3F-driven 3D parcel/scenario objects
- color-reactive glow and gradient system
- performance tiers for desktop/mid/mobile
- lazy loading for heavy scenes
- mobile fallback strategy

Build component architecture, hooks, scenes, and supporting asset structure
exactly around the LandOS Atlas cinematic implementation pack.

Preserve enterprise trust while delivering a world-class cinematic product-page
experience.
```

---

20. Final Founder Guidance

The cinematic layer should make people feel: - this is bigger than a normal SaaS - this is i is intelligent - this is inevitable - this is premium enough for serious buyers

That means every motion choice must serve story, hierarchy, and confidence.

No empty gimmicks. No random glow spam. No chaotic Web 3 aesthetics.

Only purposeful cinematic intelligence.

---

21. Final Closing Note

This pack completes the next implementation layer for LandOS Atlas cinematic storytelling.

You now have: - the cinematic experience system - the cinematic implementation pack - exact ro - exact section sequence - exact GSAP scene timelines - exact R3F object inventory - exact as brief - exact Codex build prompt

This is enough to direct Codex and design/3D agents toward a serious flagship immersive experience.